Novelty statement

Some occupancy models have been developed to deal with automated outputs but previous approaches were limited to outputs of AI classification algorithms, not differentiating multiple objects within an image. Regarding abundance models, to the best of our knowledge, no approach has been developed to account for false-negative and false-positive errors in automated counts from aerial or satellite images. Importantly, we showcase the potential of our models to more efficient automated pipelines in case studies where we estimate site occupancy of white-tail deer from camera trap images, and assess time-series abundance of Everglades wading birds from drone images.

Cover letter

Dear editors,

We are submitting our manuscript "*Estimating occupancy and abundance using automated object detections in images*" as a "**Method**" article in Ecology Letters.

We introduce two hierarchical models – one for occupancy and one for abundance estimation – that are tailored to deal with outputs of automated algorithms that detect multiple objects in remotely sensed images (e.g., camera traps and drones). Our models explicitly account for false-negative (FN) and false-positive (FP) algorithm errors using a probabilistic approach based on the output confidence scores that these algorithms provide. We evaluate the performance of our FP-FN models using simulated data and test their application in real-case examples. Specifically, we estimate occupancy of white tail deer from automated detections in camera trap images and estimate time-series abundance of Everglades bird colonies from automated counts in drone images. Interestingly, we found substantial variations in FP/FN errors in our case studies, even with finely trained/tuned object-detection algorithms, highlighting the importance of modeling spatiotemporal variations in algorithm recall (FN) and precision (FP) in automated wildlife surveys. Although some statistical models have been developed to deal with automated outputs, previous approaches were mainly limited to outputs of classification algorithms (i.e., do not differentiate individuals or species within an image). Because object-detection algorithms have become the state-of-the-art approach for detecting wildlife in images, we believe our developed frameworks can have a strong impact on ecological analysis pipelines, ultimately contributing to more efficient and reliable wildlife monitoring. Additionally, because the manuscript presents two complementary modeling approaches, it exceeds the standard length limits (approximately 6,300 words); we have asked whether this length was acceptable in the proposal letter.

Sincerely,

Ismael Verrastro Brack, on behalf of the co-authors

TITLE PAGE

Estimating wildlife occupancy and abundance using automated object detections in images

Ismael V. Brack [1*]

Sebastian Botero-Cañola [2]

S. K. Morgan Ernest [2]

Lindsey A. Garner [2]

Madison L. Heisey [2]

Ben G. Weinstein [2]

Ethan P. White [2]

Denis Valle[1]

1 – School of Forest, Fisheries and Geomatics Sciences, University of Florida, Gainesville, FL, USA

2 – Department of Wildlife Ecology and Conservation, University of Florida, Gainesville, FL, USA

***** Corresponding author: ismaelbrack@hotmail.com; *Phone:* +1 236 8865659. *Address*: 305-2469 Cornwall Ave, Vancouver, BC, Canada. V6K1B9.

Short title: Hierarchical models for automated wildlife detections

Article type: Method

keywords: automated detections, artificial intelligence, camera traps, computer vision, deep learning, drones, false positive, hierarchical model, population surveys

Number of words in abstract: 195

Number of words in main text: 6215

Number of references: 77

Number of tables: 0

Number of figures: 5

Author Contributions

IVB and DV conceived the idea, developed the statistical models, conducted data analysis, and led the manuscript writing. SBC, SKME, LAG, MLH, BGW, and EPW designed the case studies, and collected, compiled and organized the data. BGW, SKME, and EPW developed the algorithm for bird automated detections. All authors discussed results, contributed substantially to the drafts and gave final approval for publication.

Data availability statement

Data and code are available at the GitHub repository https://github.com/ismaelvbrack/Models_AI-outputs. Upon acceptance, the repository will be released and permanently archived on Zenodo with an associated Digital Object Identifier (DOI).